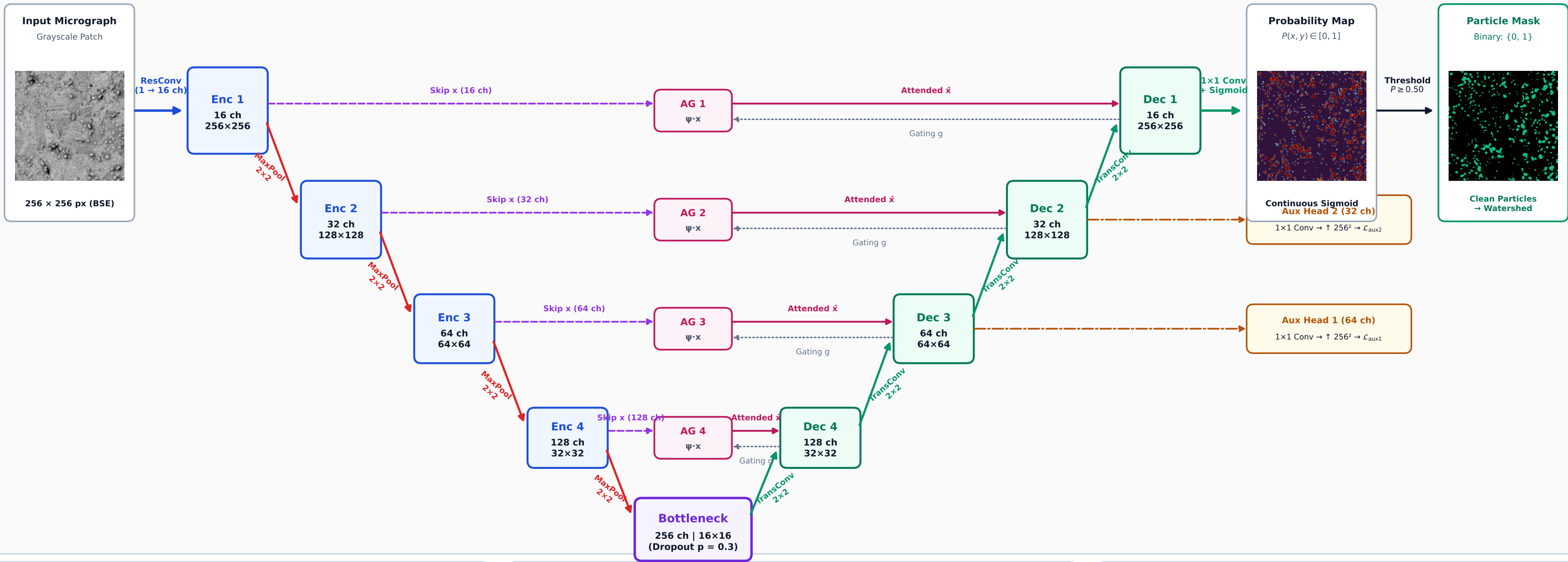
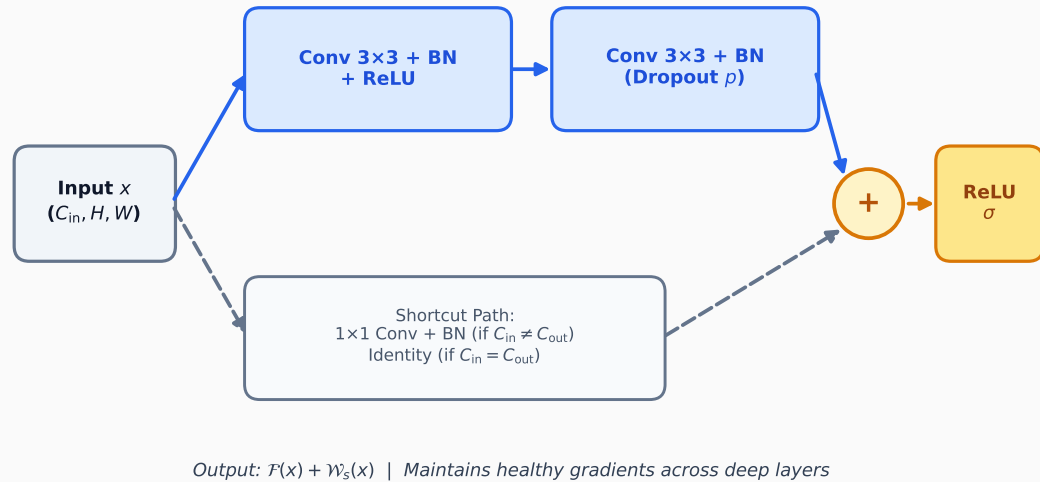


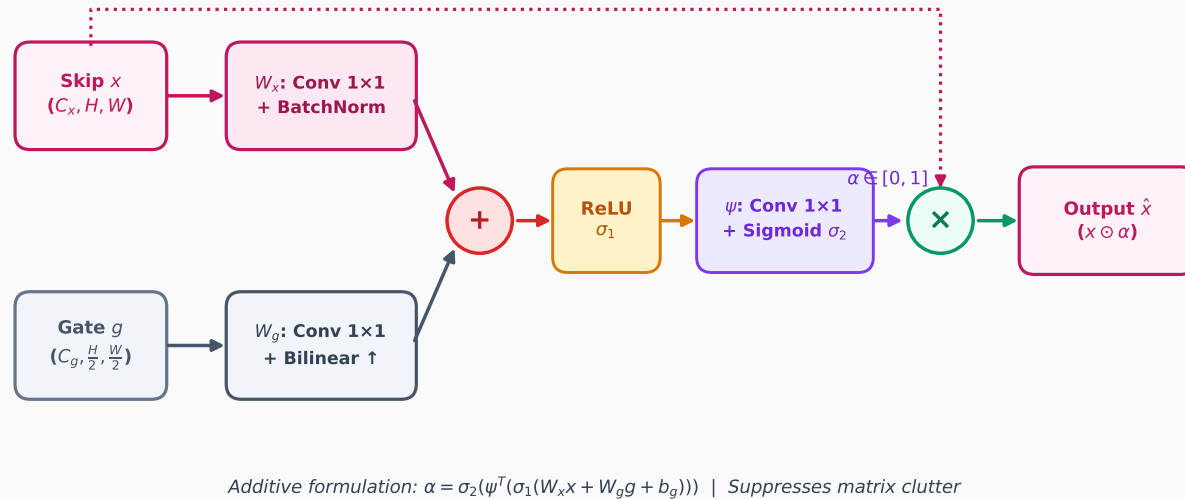
(a) Residual Attention U-Net Macro Architecture (SEM Particle Segmentation)



(b) Residual Convolutional Block (ResConvBlock)



(c) Additive Attention Gate (Oktay et al., 2018)



(d) Operations Key & Training Specifications

Operations Key	Model Specifications
ResConvBlock (Encoder Stage) 3×3 Conv + BN + ReLU + Residual Shortcut	Total Parameters: 2,053,283 (trainable)
Bottleneck Block 256 ch, Spatial Dropout $p = 0.3$	Input Size: 1 × 256 × 256 px
ResConvBlock (Decoder Stage) Concatenation + ResConv + Dropout 0.1	Channel Depths: [16, 32, 64, 128, 256]
Additive Attention Gate (AG) Filters non-particle skip activations	Combined Loss: $0.5\mathcal{L}_{BCE} + 0.5\mathcal{L}_{Dice} + 0.3\sum \mathcal{L}_{aux}$
Auxiliary Head (Deep Superv.) 1×1 Conv → Bilinear $\uparrow$ → Loss weight 0.3	Optimizer: AdamW ($\text{lr} = 10^{-3}$, $\text{wd} = 10^{-4}$) Scheduler: ReduceLROnPlateau (factor 0.5) Batch Size: 8 (170 iterations / epoch) Best Val Dice: 0.8756 (IoU = 0.7835 at Ep 24)